we find a stipulation that they apply one way in time but not in the other. Nowhere is there any distinction between how the laws look or behave when applied in either direction in time. The laws treat what we call past and future on a completely equal footing. Even though experience reveals over and over again that there is an arrow of how events unfold in time, this arrow seems not to be found in the fundamental laws of physics.

Past, Future, and the Fundamental Laws of Physics

How can this be? Do the laws of physics provide no underpinning that distinguishes past from future? How can there be no law of physics explaining that events unfold in *this* order but never in reverse?

The situation is even more puzzling. The known laws of physics actually declare—contrary to our lifetime of experiences—that light coffee can separate into black coffee and white cream; a splattered yolk and a collection of smashed shell pieces can gather themselves together and form a perfectly smooth unbroken egg; the melted ice in a glass of room-temperature water can fuse back together into cubes of ice; the gas released when you open your soda can rush back into the bottle. All the physical laws that we hold dear fully support what is known as *time-reversalsymmetry.* This is the statement that if some sequence of events can unfold in one temporal order (cream and coffee mix, eggs break, gas rushes outward) then these events can also unfold in reverse (cream and coffee unmix, eggs unbreak, gas rushes inward). I'll elaborate on this shortly, but the one-sentence summary is that not only do known laws fail to tell us why we see events unfold in only one order, they also tell us that, in theory, events can unfold in reverse order.[11]

The burning question is *Why don't we ever see such things?* I think it's a safe bet that no one has ever actually witnessed a splattered egg unsplattering. But if the laws of physics allow it, and if, moreover, those laws treat splattering and unsplattering equally, why does one never happen while the other does?

Time-Reversal Symmetry